Workflow and architecture patterns for working together 2: Architectures

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May 29, 2026

Outline

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A naive probability theory for merge consistency: isolated commits

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A naive probability theory for merge consistency: modeling branches as sets of commits

- $A = A_1 \sqcup A_2 \sqcup A_3$ where $A_1 = \{a\}$, $A_2 = \{b, c\}$, $A_3 = \{d, e\}$

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A naive probability theory for merge consistency: general definition for distinct branches diverging from a common ancestor

- Let A be the set of unmerged commits across k distinct branches $A_1 \cdots A_k$, where $A = A_1 \sqcup A_2 \sqcup \cdots \sqcup A_k$, $n = |A|$, and $n_i = |A_i|$ for $1 \leq i \leq k$. Then

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$$P(no\ conflicts) = \frac{1}{2^n - (\sum_{i=1}^{k} 2^{n_i}) + k}$$

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The State of DevOps Report: throughput metrics

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The State of DevOps Report: stability metrics

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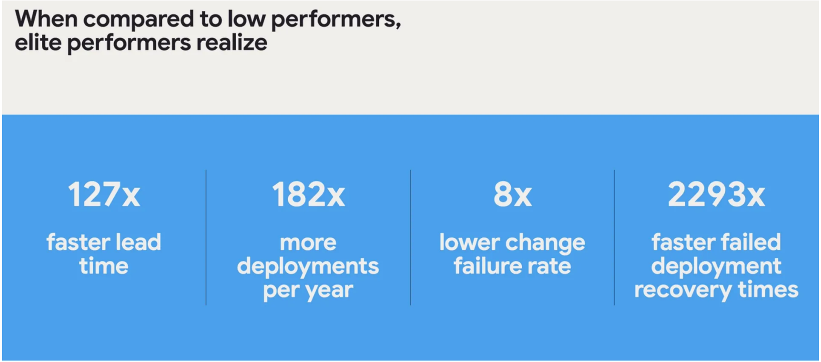
The State of DevOps Report: results

'Astonishingly, these results demonstrate that there is no trade-off between improving performance and achieving higher levels of stability and quality. Rather, high performers do better at all of these measures. This is precisely what the Agile and Lean movements predict, but much dogma in our industry still rests on the false assumption that moving faster means trading off against other performance goals, rather than enabling and reinforcing them.' - Nicole Forsgren, Jez Humble, and Gene Kim, Accelerate (2017), p. 14

The State of DevOps Report: results



- 2024 State of DevOps Report

Approximating DevOps metrics in our current environment

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Throughput: overall velocity metrics

Branch	Runs	Earliest run	Contributors
NG_Claims_837D_COB	74	23 Mar	5
PEDI-35120_837D_COB	85	15 Apr	5
Either COB branch	159	23 Mar	6
All other branches	123	23 Mar	12
	72	15 Apr	10
	266	21 Jan	

Source: GH Actions Simulation tests, 28 May 2026

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Stability metrics: merge work ratio

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Excluding PR merges	14/79 (18%)

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- 50% less merge work
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- 50% less merge work
- 25% less backmerge work
- Source command: `git log --since="2026-04-15" --oneline <--merges> <--all> <^PEDI-35120_837D_COB> | <grep -iv ''merge pull request''> | wc -l`

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- these teams are *stream-aligned*, i.e., they focus on a particular business area (in our case, either a transaction type or a closely related set of transaction types)
- these teams are *autonomous*, i.e., they can move work forward without requiring outside assistance from another team

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- all teams working in same git repository are in the same daily scrum
- each team in the above sense has its own Microsoft Teams chat with only its own contributors in it
- these teams are *stream-aligned*, i.e., they focus on a particular business area (in our case, either a transaction type or a closely related set of transaction types)
- these teams are *autonomous*, i.e., they can move work forward without requiring outside assistance from another team

Team composition: enabling teams

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Outline

1. Current challenges and their canonical solutions
2. Branching: theory and statistics
   - Theory
   - Statistics
     - Devops Statistics recapitulation
     - Approximating DevOps metrics in our current setup
     - Velocity
     - Safety
     - Maintenance to value-adding work percentage
3. Going forward
   - Team Composition
   - Branching and committing
   - Testing
   - Pull Requests
   - Pipelines
4. Architecture

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Outline

- Every change is tested by the test suite locally before committing

- tests are written against new code as that code is being written

Outline

- daily PRs - stacked PRs

Outline

- pipeline is definitive for deployment - daily stale branch job

Outline

Outline

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